LibVerse

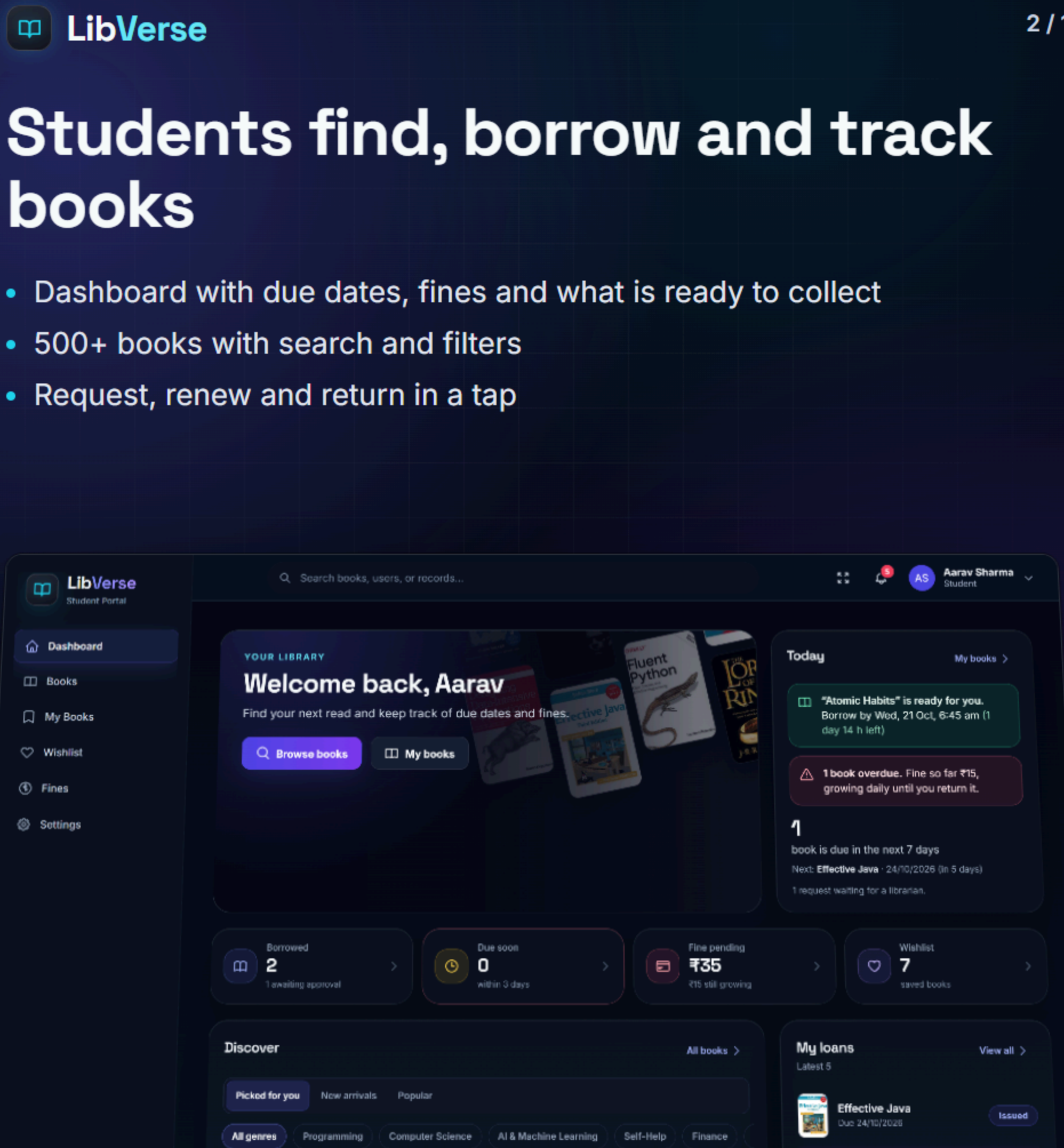
Smart Library Management System

One app for students, librarians and admins: borrowing, fines, reservations, QR cards and AI picks.



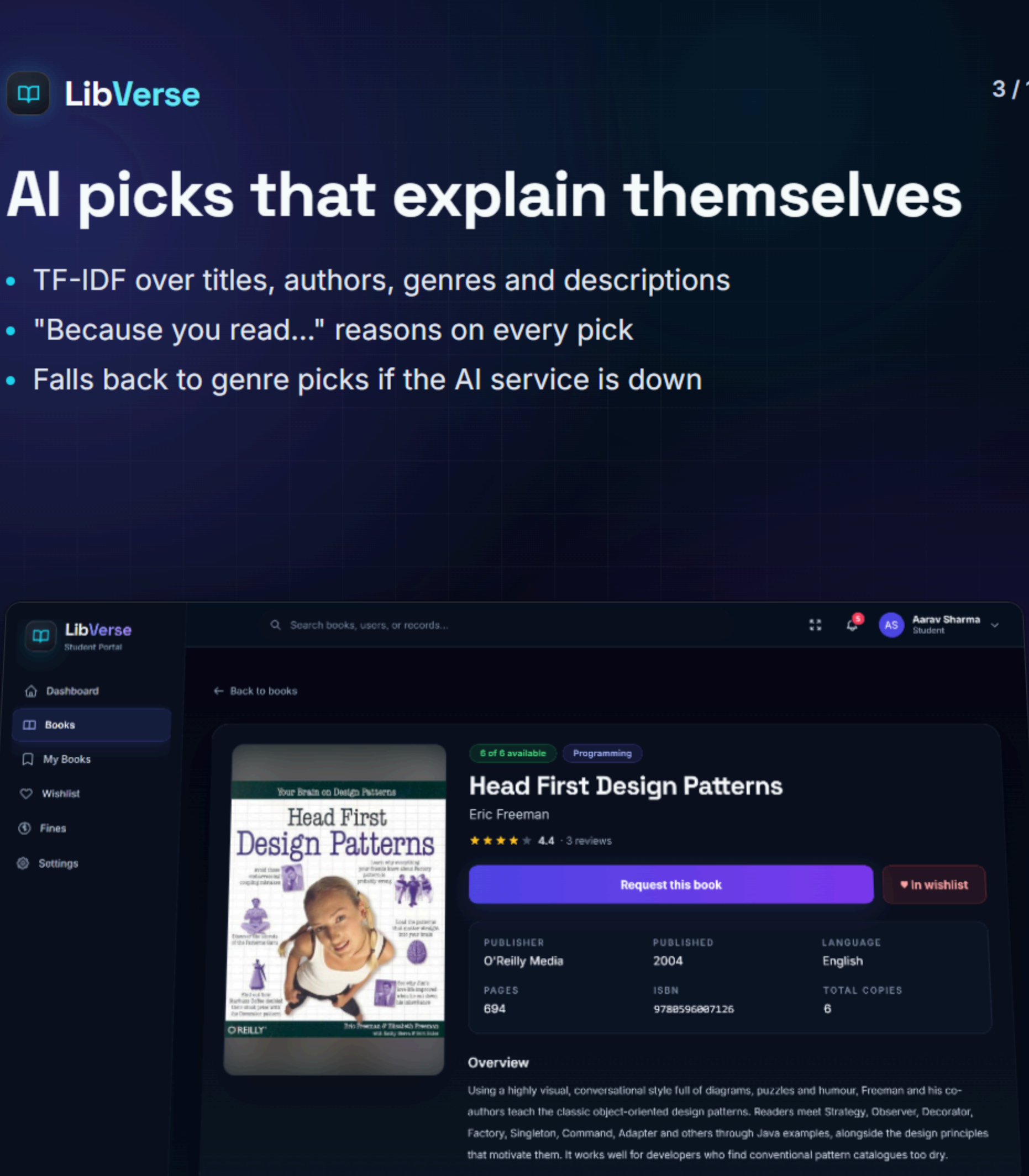
Students find, borrow and track books

- Dashboard with due dates, fines and what is ready to collect
- 500+ books with search and filters
- Request, renew and return in a tap



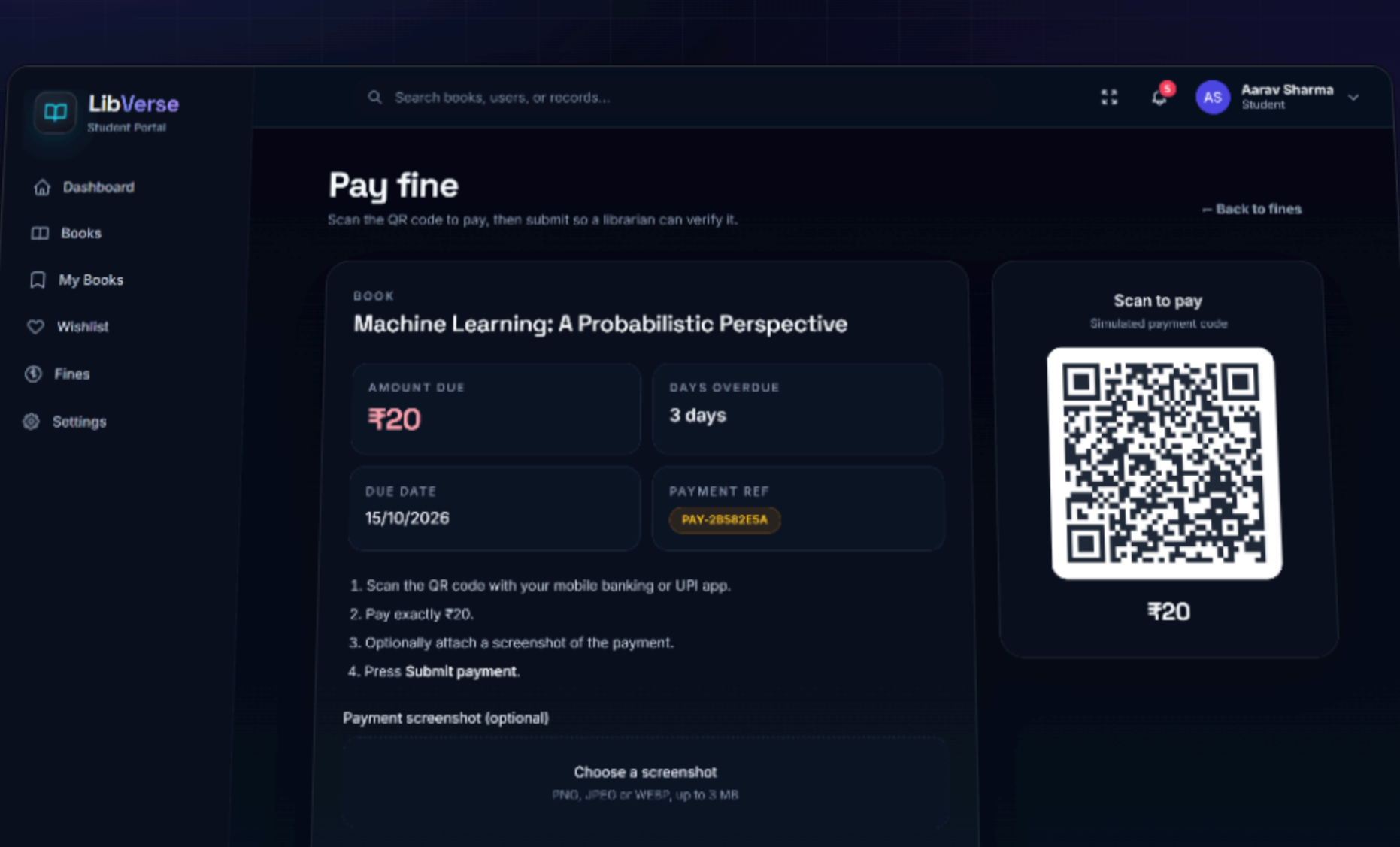
AI picks that explain themselves

- TF-IDF over titles, authors, genres and descriptions
- "Because you read..." reasons on every pick
- Falls back to genre picks if the AI service is down



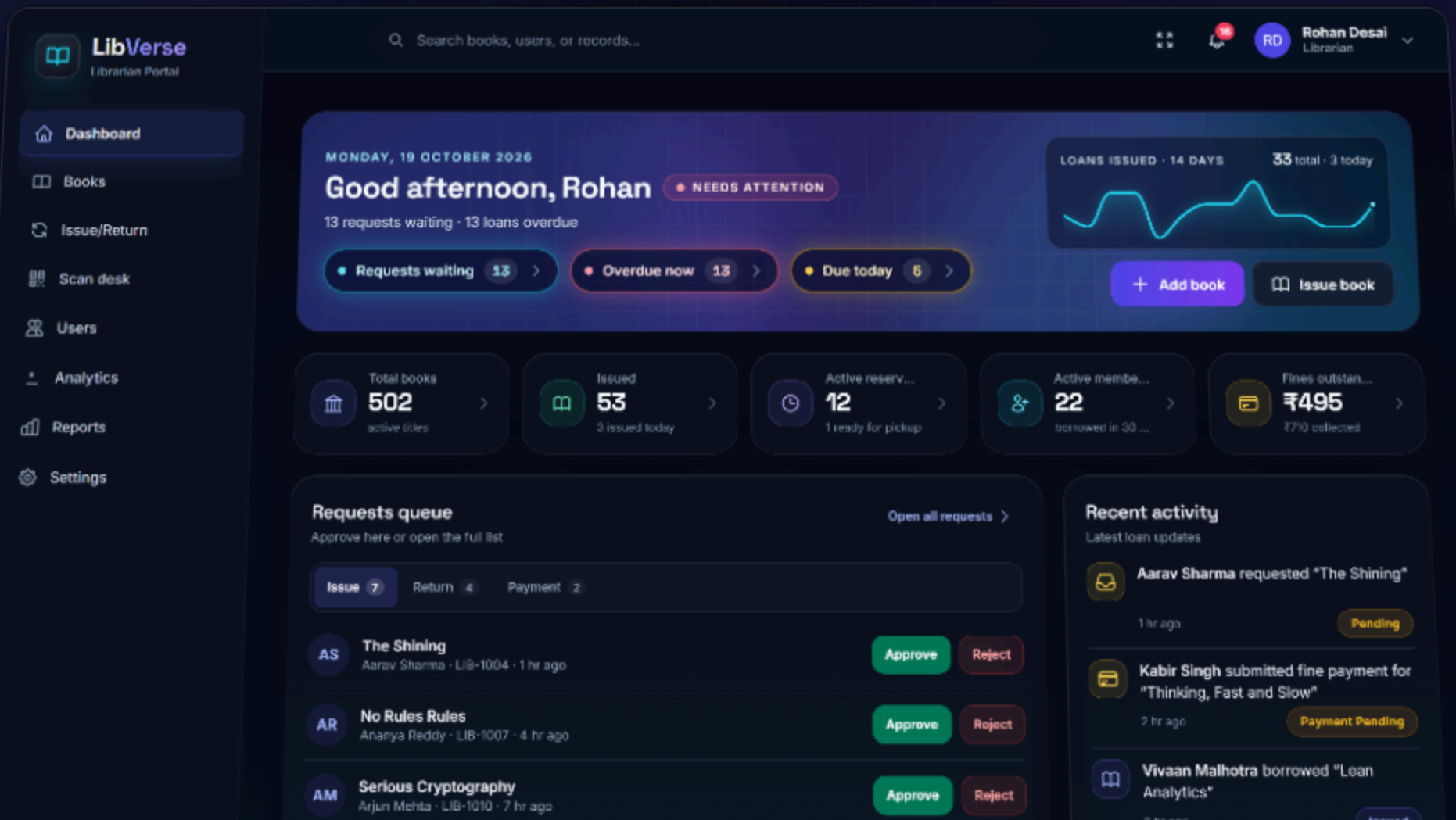
Fines without the paperwork

- ₹5 a day, worked out automatically
- Pay by UPI QR and upload the receipt
- Librarians verify or waive with a reason



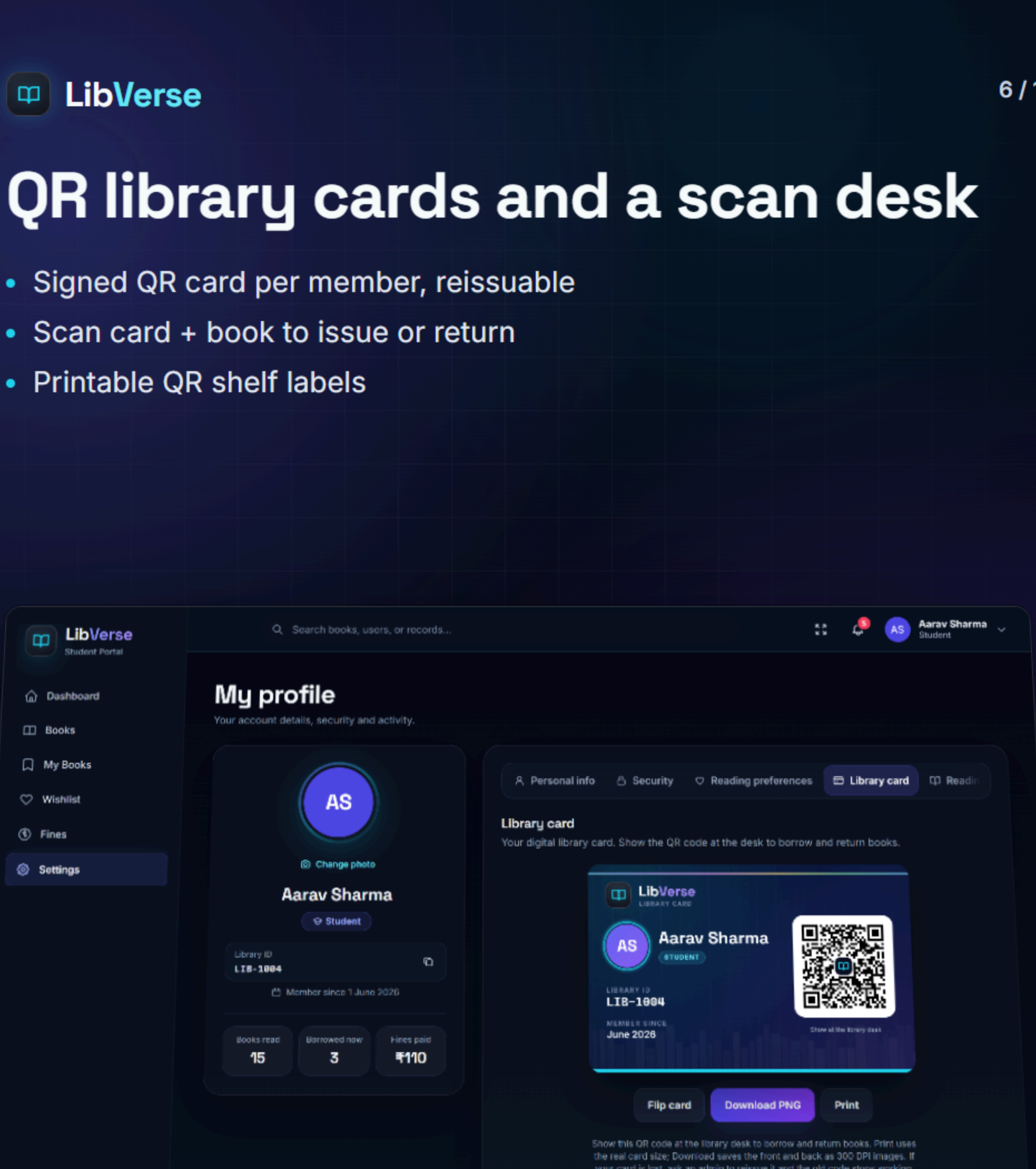
A control room for librarians

- Requests, overdue and due-today at a glance
- Approve issues and returns in one place
- 14-day activity trend



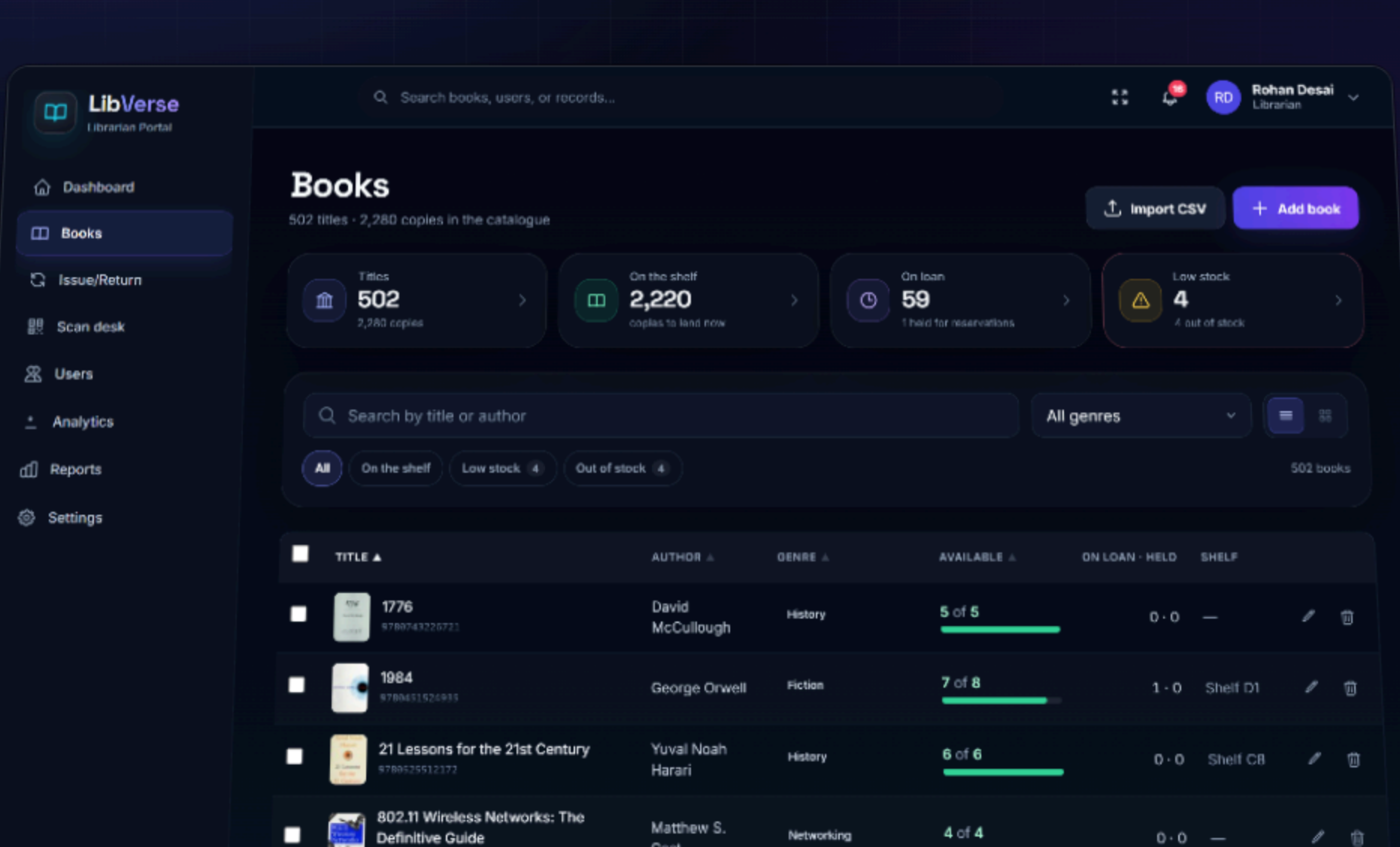
QR library cards and a scan desk

- Signed QR card per member, reissuable
- Scan card + book to issue or return
- Printable QR shelf labels



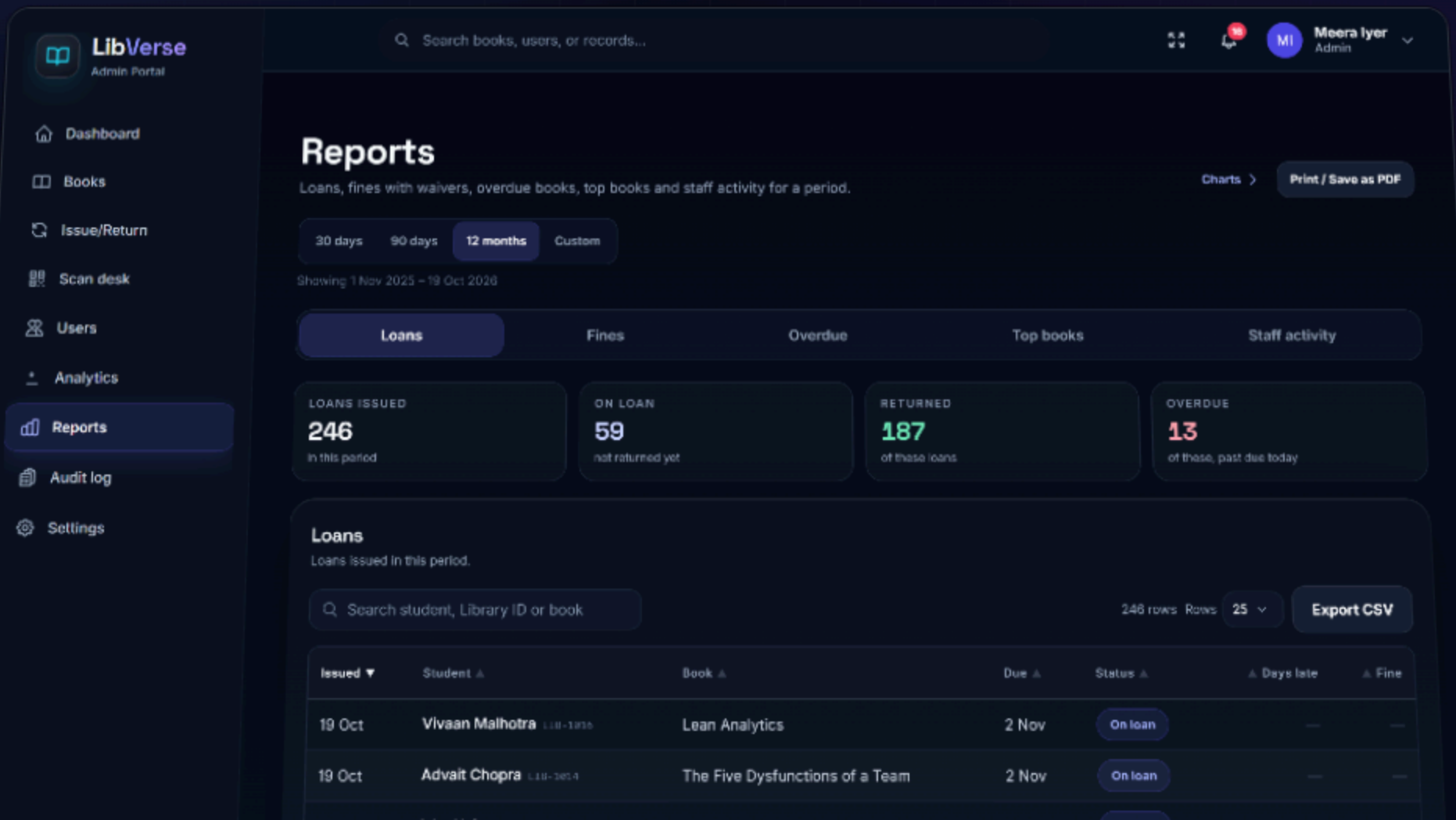
Catalogue tools that scale

- Sortable table with stock at a glance
- Low-stock filter and bulk actions
- CSV import with a row-by-row preview



Reports, analytics and an audit log

- Loans, fines, overdue and top books
- Staff activity from the audit log
- CSV export and print



LibVerse

Tech stack

Frontend
React 18, Vite, Tailwind CSS, ECharts, code-split routes

Backend
Node.js, Express, MongoDB with Mongoose, GridFS uploads

AI service
Python, Flask, scikit-learn TF-IDF with explanations

Security
helmet CSP, rate limits, JWT sessions ended on password change

Quality
Jest + Supertest, pytest, GitHub Actions CI

Deploy
Vercel, Render, MongoDB Atlas

LibVerse

Thanks for reading

LibVerse is open source. The code, tests, screenshots and a step-by-step deployment guide are on GitHub (link in the post).

React Node.js MongoDB Python